SENIOR PROJECT BL1-2026

Software for Attendance Check for Staff Using Geofencing Technology

Submitted to

School of Information, Computer and Communication Technology
Sirindhorn International Institute of Technology
Thammasat University

Submitted date 10/03

By

Thanyawach Keawhan 6622780094
Nontapat Chaopratoom 6622772489
Suleyman Hannyyev 6622790291

Advisor: DR. BING GWONG LEUNG
DR. SOMRUDEE DEEPAISARN

Software for Attendance Check for Staff Using Geofencing Technology

Abstract

Attendance management is an important administrative process for educational institutions. Conventional attendance systems often require staff members to manually record their arrival and departure using methods such as attendance sheets, identification cards, biometric scanners, QR codes, or attendance applications. These methods depend on staff members remembering to perform the required action and may result in missing or inaccurate attendance records when check-in or check-out is forgotten.

This project proposes a **software-based attendance system using geofencing technology** for professors, faculty staff, and other authorized personnel. A virtual geographic boundary, known as a geofence, is established around the faculty building. When a registered staff member enters the defined faculty area, the mobile application detects the location transition and displays a notification requesting the staff member to manually check in. Similarly, when the staff member leaves the faculty area, the application displays a notification requesting manual check-out.

To reduce missing attendance records, the proposed system also includes an automatic attendance mechanism. If a check-in notification remains unanswered for a predefined period, the system checks the staff member's latest location again. If the staff member is still located within the faculty area, the system automatically records the check-in. For check-out, if the notification remains unanswered and the staff member is confirmed to be outside the faculty area after the predefined period, the system automatically records the check-out.

The proposed system combines manual confirmation with location-based automation. Manual confirmation remains the preferred attendance method, while automatic check-in and check-out act as backup mechanisms when staff members forget to respond. The system will be evaluated based on geofence detection accuracy, attendance-recording accuracy, detection delay, notification delay, and overall system reliability in real-world conditions.

Introduction

Attendance records are commonly used by organizations and educational institutions to manage working hours, maintain personnel records, and support administrative processes. In universities, professors and faculty staff may be required to record when they arrive at and leave their workplace.

Traditional attendance systems can use paper attendance sheets, employee identification cards, fingerprint scanners, facial recognition, QR codes, or dedicated attendance terminals. Although these systems can provide attendance records, most require users to actively interact with a device or application whenever they arrive at or leave the workplace.

This creates a practical problem when staff members forget to record their attendance. For example, a professor may arrive at the faculty building and immediately begin teaching or working without remembering to check in. Similarly, a staff member may leave the faculty building without checking out. Missing attendance records may later require manual correction and additional communication between the staff member and administrative personnel.

Modern smartphones provide an opportunity to reduce this problem because they contain location technologies such as GPS, Wi-Fi positioning, and mobile network positioning. Mobile operating systems also provide **geofencing**, which allows an application to monitor whether a mobile device enters or exits a predefined geographic area.

Android's official geofencing system supports events for entering, exiting, and dwelling within a defined geographic area. However, geofence notifications are not guaranteed to occur immediately, particularly when applications operate in the background. Location accuracy may also vary according to environmental and network conditions.

Research has previously investigated the use of geofencing for attendance management. Eweoya et al. developed a university attendance management system using geofencing to automate attendance monitoring at Babcock University. Their work demonstrated the potential of geofencing for reducing manual attendance processes in educational environments.

Another IEEE conference study proposed an attendance management system combining GPS and geofencing as an alternative to traditional paper-based and hardware-dependent attendance methods.

The proposed project applies the concept specifically to **faculty employees, professors, and other staff members**. Unlike a completely automatic location-based attendance system, this project uses a **hybrid manual and automatic approach**. Crossing the geographic boundary first produces a notification asking the user to confirm their attendance. Automatic attendance recording occurs only if the user does not respond within a predefined period and their location is verified again.

Problem Statement

Existing attendance systems often require employees to actively remember to check in when arriving at the workplace and check out when leaving. When employees forget to perform these actions, attendance records may become incomplete or inaccurate.

A system that automatically records attendance immediately after detecting a location change may solve the problem of forgotten check-ins, but it introduces another problem. Smartphone location information is not perfectly accurate. A device located close to the geographic boundary may occasionally be detected on the wrong side of the boundary.

Therefore, automatically recording attendance based on only one location event could produce incorrect attendance records.

This project addresses these problems using a hybrid attendance mechanism consisting of:

1. **Location-based detection** to detect when staff members enter or leave the faculty area.
2. **Manual confirmation** through a mobile notification as the primary method of recording attendance.
3. **Automatic check-in or check-out** if the user does not respond to the notification within a predefined period.
4. **Location revalidation** before any automatic attendance action is performed.

Importance of the Project

The proposed system aims to make staff attendance recording more convenient without completely removing user control.

Instead of requiring staff members to remember to open an application every day, the system automatically detects when attendance may need to be recorded and sends the appropriate notification.

At the same time, the system does not immediately assume that every detected geographic transition should become an attendance record. Staff members are first given an opportunity to manually confirm the action.

The automatic mechanism serves as a backup if the staff member forgets to respond.

Impact and Benefits

The proposed system may provide the following benefits:

- Reduce forgotten staff check-ins.
- Reduce forgotten staff check-outs.
- Reduce incomplete attendance records.
- Reduce administrative work required to correct missing attendance records.
- Automatically remind staff when they arrive at the faculty.
- Automatically remind staff when they leave the faculty.
- Allow attendance to be recorded using smartphones that staff already carry.
- Reduce dependence on dedicated attendance hardware.
- Maintain digital and timestamped attendance records.
- Distinguish between manually confirmed and automatically generated attendance.
- Allow administrators to review staff attendance from a centralized system.
- Provide a more convenient attendance experience for faculty employees.
- Provide information for evaluating the practical reliability of smartphone geofencing around university buildings.

Framework

The proposed attendance system consists of several major components:

1. Staff Mobile Application
2. Geofencing and Location Module
3. Notification Module
4. Attendance Decision Module
5. Backend Server
6. Attendance Database
7. Administrator Interface

The system establishes a virtual geographic boundary around the faculty building. The staff member's mobile device monitors transitions into and out of this area.

Overall System Framework

The general process of the system is:

Step 1: Staff smartphone monitors location.

Step 2: The location and geofencing service detects whether the staff member enters or

leaves the faculty area.

Step 3: The notification system displays a check-in or check-out request.

Step 4: The system waits for manual confirmation.

Step 5: If the user does not respond before the timeout, the system revalidates the location.

Step 6: The attendance decision module determines whether an automatic check-in or check-out should occur.

Step 7: The backend server processes the attendance record.

Step 8: The attendance database stores the result.

Step 9: The administrator dashboard displays the attendance information.

Check-In Framework

The check-in procedure begins when a staff member enters the faculty area.

Check-In Flow

Step 1: Detect Entry

The staff member approaches and enters the faculty building.

The mobile phone detects that the device has entered the faculty geofence.

Step 2: Send Notification

The system displays a notification asking the staff member to check in.

Step 3: Check for Manual Response

Question:

Did the staff member press "Check In"?

If YES:

1. Record manual check-in.
2. Save the check-in time.
3. Mark the attendance method as "Manual."
4. Send the attendance record to the database.
5. Finish the check-in process.

If NO:

1. Keep the check-in request active.
2. Start the grace-period timer.
3. Wait until the grace period expires.
4. Request or verify the latest staff location.

The system then checks:

Is the staff member still inside the faculty geofence?

If YES:

1. Automatically record the check-in.
2. Save the check-in time.
3. Mark the attendance method as "Automatic."
4. Send the attendance record to the database.
5. Finish the check-in process.

If NO:

1. Do not record a check-in.
2. Cancel or expire the pending check-in request.
3. Finish the check-in process.

Check-Out Framework

The check-out procedure begins when the staff member leaves the faculty area.

Check-Out Flow

Step 1: Detect Exit

The staff member leaves the faculty building.

The mobile phone detects that the device has exited the faculty geofence.

Step 2: Send Notification

The system displays a notification asking the staff member to check out.

Step 3: Check for Manual Response

Question:

Did the staff member press "Check Out"?

If YES:

1. Record manual check-out.
2. Save the check-out time.
3. Mark the attendance method as "Manual."
4. Send the attendance record to the database.
5. Finish the check-out process.

If NO:

1. Keep the check-out request active.
2. Start the grace-period timer.
3. Wait until the grace period expires.
4. Request or verify the latest staff location.

The system then checks:

Is the staff member still outside the faculty geofence?

If YES:

1. Automatically record the check-out.
2. Save the check-out time.
3. Mark the attendance method as "Automatic."
4. Send the attendance record to the database.
5. Finish the check-out process.

If NO:

1. Do not automatically check out.
2. Cancel the pending automatic check-out.
3. Continue monitoring the staff member's location.

Automatic Check-In Decision Rule

The automatic check-in rule can be written simply as:

Auto Check-In = YES

when:

Notification Timeout Reached AND Staff Location = Inside Faculty Geofence

Otherwise:

Auto Check-In = NO

The possible cases are:

- Timeout reached and staff is still inside = Automatic Check-In
- Timeout reached and staff is outside = No Check-In
- Staff manually checks in before timeout = Manual Check-In
- Timeout has not been reached = Continue waiting

Automatic Check-Out Decision Rule

The automatic check-out rule is:

Auto Check-Out = YES

when:

Notification Timeout Reached AND Staff Location = Outside Faculty Geofence

Otherwise:

Auto Check-Out = NO

The possible cases are:

- Timeout reached and staff is outside = Automatic Check-Out
- Timeout reached and staff is back inside = No Automatic Check-Out
- Staff manually checks out before timeout = Manual Check-Out
- Timeout has not been reached = Continue waiting

Chapter 2: Proposed System and Functional Specifications

2.1 System Overview

The proposed project, Software for Attendance Check for Staff Using Geofencing Technology, is a mobile-based attendance system for professors, faculty staff, and other authorized personnel.

The system uses the smartphone's location information to determine whether the device is inside or outside a predefined institutional region. When an eligible entry or exit is detected, the application displays a notification requesting manual check-in or check-out.

Manual confirmation is the primary attendance method. If the user does not respond within the configured grace period, the system requests a fresh location reading and evaluates whether automatic attendance recording is appropriate.

The system consists of a staff mobile application, a location and geofencing module, a notification module, a backend server, a centralized database, and an administrator web interface.

This chapter defines the proposed system's scope, user roles, authentication process, interface design, location rules, attendance workflow, and database structure. These specifications describe the intended implementation rather than completed or experimentally verified functions.

2.2 System Scope and Initial Configuration

2.2.1 Institutional Region

The institutional region is defined as a circular area around a selected reference point at the faculty. The reference point consists of a latitude and longitude configured by an administrator.

For the initial prototype, the proposed radius is 200 metres. A device within this distance is geometrically inside the institutional region; a device farther away is geometrically outside.

The radius is an attendance-policy boundary and does not represent the exact outline of the faculty building. Therefore, being inside the region does not necessarily establish that a

person is inside a particular room or building.

The prototype will support a configurable positive radius of up to 1,000 metres. The final radius must be selected after reviewing the faculty layout and conducting field tests.

2.2.2 Initial Prototype Parameters

Parameter	**Proposed initial specification**
Institutional reference point	Latitude and longitude selected at the faculty site
Institutional radius	200 metres
Maximum supported radius for this local-area design	1,000 metres
Manual-response grace period	5 minutes
Maximum age of a location reading used for attendance approval	60 seconds
Maximum reported horizontal uncertainty accepted for attendance approval	30 metres
User roles	Staff Member and Administrator
Attendance methods	Manual and Automatic
Attendance storage	Centralized backend database
Display time zone	Asia/Bangkok

These values are initial design parameters. They must be evaluated during implementation and adjusted if testing shows that they do not provide an appropriate balance between usability and attendance reliability.

2.2.3 Scope Limitations

The prototype will record attendance sessions within one configured institutional region. It will support more than one attendance session per day, provided that each user has no more than one open session at a time.

The project does not initially include payroll calculation, leave approval, biometric identity verification, or room-level positioning.

Location-based attendance indicates the position reported by a registered device. It does not independently prove that the account holder is physically carrying the device or performing work.

2.3 User Roles and Access Permissions

2.3.1 Staff Member

The Staff Member role includes professors, faculty employees, and other authorized personnel.

Staff members will be able to:

1. Log in using an authorized account.
2. View their current attendance status.
3. Receive check-in and check-out notifications.
4. Manually confirm attendance.
5. View their own attendance history.
6. See whether attendance was recorded manually or automatically.
7. View permission, location, and synchronization warnings.
8. Submit a correction request for an incorrect or missing attendance record.
9. Log out of the application.

Staff members will not be permitted to view another user's attendance history, change geofence settings, assign roles, or directly modify stored attendance records.

2.3.2 Administrator

Administrators will manage the system through a protected web interface.

Administrators will be able to:

1. Create, activate, and deactivate staff accounts.
2. Assign authorized user roles.
3. View and search attendance records.
4. Filter attendance by user, date, and recording method.
5. Review open sessions and unresolved attendance issues.
6. Review and approve or reject correction requests.

7. Configure the institutional centre, radius, and grace period.
8. Export attendance reports.
9. Review system events and audit logs.

Every administrative attendance correction must include a reason. The system must preserve the original values and record who made the correction and when.

2.3.3 Permission Enforcement

The backend server will enforce access permissions for every protected request.

Hiding a button in the interface will not be treated as sufficient access control. For example, a staff account requesting another employee's attendance records must be rejected by the backend.

2.4 Login and Account Management

2.4.1 Account Information

Each account will contain:

- A unique internal user identifier.
- A unique staff identifier.
- Full name.
- Institutional email address.
- Department.
- Assigned role.
- Account status.
- Password hash.
- Account creation and update timestamps.

The proposed prototype will use institutional email and password authentication. Integration with an institutional single sign-on service is outside the initial specification and may be considered later.

2.4.2 Account Creation and Activation

An administrator will create the account using verified staff information. The staff member will then complete an activation process and set a personal password using a time-limited, single-use activation mechanism.

Public self-registration will not be included in the initial prototype.

Passwords must not be stored as readable text. The backend will store a salted password hash and will not expose it through the mobile application or administrator interface.

2.4.3 Login Procedure

The proposed login procedure is:

1. The user enters an institutional email address and password.
2. The application submits the credentials through an encrypted connection.
3. The backend verifies the credentials and account status.
4. If authentication succeeds, the backend establishes an authenticated session.
5. The application displays the interface appropriate to the user's role.
6. If authentication fails, the application displays a general error message without revealing whether the email address exists.

Repeated failed login attempts will be restricted. Password reset will use a time-limited, single-use mechanism.

2.4.4 Session and Logout Behaviour

Authentication credentials used to maintain a session will be stored using the device's protected storage facilities.

Logging out will end the application's authenticated session and stop attendance monitoring for that account on the device. Logging out will not automatically check the user out of an existing attendance session.

If the user has an open attendance session, the application will warn that attendance monitoring will stop. The existing session will remain available for later completion or administrative review.

Deactivated accounts will not be allowed to submit new attendance actions.

2.5 User Experience and Interface Design

2.5.1 Login Screen

The login screen will contain:

- Institutional email field.
- Password field.
- Log In button.
- Forgot Password option.
- Authentication error message area.

After the first successful login, the application will explain how attendance monitoring works, what permissions it needs, and when automatic attendance may occur.

2.5.2 Permission and Setup Screen

The setup screen will explain the purpose of location and notification permissions and display whether each required permission is enabled.

If a required permission is unavailable, the application will clearly identify the affected feature. It must not display attendance monitoring as active when it cannot obtain the information required to operate.

2.5.3 Staff Dashboard

The staff dashboard will display:

- Staff name and identifier.
- Current attendance state.
- Current location classification: Inside, Outside, or Uncertain.
- Latest location update time.
- Current check-in and check-out information.
- Pending attendance request and remaining grace period.
- Available manual attendance action.
- Synchronization status.
- Access to attendance history and correction requests.

Attendance and location will be displayed separately. For example, a user may be “Checked In” while the current location classification is “Uncertain.”

An action will be shown as successfully recorded only after the backend confirms it. An action awaiting upload will be labelled “Pending Synchronization.”

2.5.4 Check-In Notification

A proposed check-in notification is:

“You appear to be within the faculty attendance area. Please confirm your check-in. If you do not respond within 5 minutes, the system will attempt automatic check-in after verifying your location again.”

The notification will provide Check In and Dismiss actions.

Dismissing the notification will close the visible notification but will not cancel the pending attendance request or its grace period. This behaviour will be explained during setup and shown on the dashboard.

2.5.5 Check-Out Notification

A proposed check-out notification is:

“You appear to have left the faculty attendance area. Please confirm your check-out. If you do not respond within 5 minutes, the system will attempt automatic check-out after verifying your location again.”

The notification will provide Check Out and Dismiss actions.

A check-out request will only be created when the user has an open attendance session.

2.5.6 Attendance History Screen

The attendance history screen will show:

- Attendance date.
- Check-in and check-out times.
- Manual or automatic method for each action.
- Open or completed session status.
- Any approved correction.
- Any unresolved issue requiring review.

Users will be able to select a date range and submit a correction request with an explanation.

2.5.7 Administrator Dashboard

The administrator dashboard will include:

- Attendance overview.
- Staff account management.
- Attendance search and filtering.

- Open-session review.
- Correction-request review.
- Institutional-region settings.
- Report export.
- Audit-log access.

Manual attendance, automatic attendance, and administrator-corrected attendance will be distinguishable.

2.6 Location Acquisition and Institutional Distance

2.6.1 Reading the Device Location

The application will obtain latitude and longitude from the smartphone's location service. The application does not need to calculate the device's position from satellite signals.

Each location reading used for an attendance decision will include:

- Latitude.
- Longitude.
- Reading timestamp.
- Reported horizontal uncertainty, when available.

The application will use the location reading to determine the device's distance from the configured institutional reference point.

2.6.2 Local Straight-Line Distance

Because the institutional region is small, the prototype will approximate the surrounding area as a flat local surface. The required measurement is the straight-line distance from the device to the institutional centre, rather than a walking route or road distance.

Latitude and longitude cannot be treated directly as distances in metres. They will first be converted into approximate local east–west and north–south displacements.

Let:

lat_0 = institutional-centre latitude in radians.
lon_0 = institutional-centre longitude in radians.
lat = device latitude in radians.
lon = device longitude in radians.
R = approximate Earth radius, 6,371,000 metres.
x = local east–west displacement in metres.
y = local north–south displacement in metres.
d = approximate straight-line distance in metres.

Convert degrees to radians using:

$$\text{Radians} = \text{Degrees} \times \pi / 180$$

Calculate:

$$x = R \times \cos(lat_0) \times (lon - lon_0)$$

$$y = R \times (lat - lat_0)$$

$$d = \sqrt{(x^2 + y^2)}$$

This approach uses the Earth-radius constant to convert angular coordinates into local distances. It does not require a curved-surface distance calculation such as Haversine for the intended small institutional region.

2.6.3 Basic Region Rule

Let r represent the configured institutional radius.

The geometric classification is:

- Inside when $d \leq r$.
- Outside when $d > r$.

For the proposed 200-metre radius, a reported position 150 metres from the centre is geometrically inside, while a position 250 metres away is geometrically outside.

These examples illustrate the rule and are not experimental measurements.

2.6.4 Location Uncertainty

The attendance decision will also consider the uncertainty associated with the location reading.

Let a represent the reported horizontal uncertainty in metres. The proposed decision rule is:

Confirmed Inside when $d + a \leq r$.
Confirmed Outside when $d - a > r$.
Uncertain when the uncertainty range overlaps the boundary.

A reading will also be classified as Uncertain if it is older than 60 seconds, has reported uncertainty greater than 30 metres, or lacks the information required for validation.

This is a conservative decision rule, not a guarantee that the true device position lies within the reported uncertainty range.

An Uncertain classification will not authorize automatic attendance. The system will request another reading or mark the action for review if validation remains unavailable.

2.6.5 Boundary Handling

A geofence transition will trigger an attendance evaluation; it will not directly become an attendance record.

The system will reduce repeated or contradictory actions by:

1. Revalidating location before recording attendance.
2. Maintaining only one active attendance request per user.
3. Cancelling a pending check-in when a confirmed exit occurs before completion.
4. Cancelling a pending check-out when a confirmed return occurs before completion.
5. Preventing duplicate processing of the same event.

A cancelled request will not be resumed automatically. A later eligible transition will create a new request with a new grace period.

2.7 Attendance States and Decision Rules

2.7.1 Attendance States

State	Meaning
Checked Out	No attendance session is currently open
Pending Check-In	An eligible entry has produced a check-in request
Checked In	An attendance session is open
Pending Check-Out	An eligible exit has produced a check-out request for the open session

Location uncertainty and synchronization status will be recorded separately from these attendance states.

2.7.2 Manual Check-In

Manual check-in will require:

1. An authenticated and active account.
2. No existing open attendance session.
3. A fresh location reading classified as Confirmed Inside.
4. Backend approval of the request.

After approval, the backend will create an attendance session and mark the check-in method as Manual.

If the location is Uncertain or Confirmed Outside, the application will explain why check-in cannot be completed and offer a retry or correction-request option.

2.7.3 Automatic Check-In

Automatic check-in will require all of the following:

1. A valid pending check-in request.
2. A successfully posted notification recorded by the application.
3. Expiry of the configured grace period.
4. No completed manual check-in for that request.
5. No existing open attendance session.
6. A fresh reading classified as Confirmed Inside.
7. Backend validation and approval.

If the user is Confirmed Outside, the pending request will be cancelled.

If the location cannot be validated, the request will be marked unresolved without creating an attendance session. Later completion will require a new validated action or administrative correction.

2.7.4 Manual Check-Out

Manual check-out will require:

1. An authenticated and active account.
2. An existing open attendance session.
3. A fresh reading classified as Confirmed Outside.
4. Backend approval.

After approval, the backend will close the session and mark the check-out method as Manual.

If the user needs to end attendance while still inside the institutional region, this will be handled through a correction request in the initial prototype.

2.7.5 Automatic Check-Out

Automatic check-out will require all of the following:

1. A valid pending check-out request.
2. A successfully posted notification recorded by the application.
3. Expiry of the configured grace period.
4. No completed manual check-out for that request.
5. An existing open attendance session.
6. A fresh reading classified as Confirmed Outside.
7. Backend validation and approval.

If the user is Confirmed Inside again, the pending check-out request will be cancelled and

the attendance session will remain open.

If the location cannot be validated, the request will be marked unresolved and the attendance session will remain open for review.

2.7.6 Timestamp Policy

The system will distinguish between:

- Geofence detection time.
- Notification posting time.
- Manual confirmation time.
- Location measurement time.
- Attendance recording time.
- Backend receipt time.

For the initial prototype, the official attendance timestamp will be the backend approval time. An automatic action will not be backdated to the original boundary crossing.

This means an automatic attendance time may be later than the physical arrival or departure time. The distinction will be documented and included in the evaluation.

2.7.7 Duplicate and Simultaneous Actions

A manual confirmation and an automatic action may arrive at nearly the same time. The backend must process them atomically so that only one valid attendance action is stored.

Each attendance request will have a unique identifier. Repeated submissions of the same request must return the existing result rather than create additional attendance records.

2.8 System Components and Responsibilities

2.8.1 Staff Mobile Application

The mobile application will provide authentication, permission guidance, attendance notifications, manual actions, attendance history, and synchronization status.

It will collect location evidence needed for attendance validation and submit requests to the backend.

2.8.2 Location and Geofencing Module

This module will monitor the configured institutional region, receive location-transition events, and obtain updated location readings.

It will supply location evidence to the attendance workflow. It will not independently approve or modify official attendance records.

2.8.3 Notification Module

The notification module will create check-in and check-out requests, record notification posting, and display pending actions in the application.

Posting a notification does not establish that the user has read it. The system will therefore distinguish between notification posting and user response.

2.8.4 Attendance Decision Module

The attendance decision module will evaluate the user's attendance state, pending request, grace-period expiry, and location evidence.

The backend will make the final decision to create or close an attendance session. Grace-period expiry alone will not be sufficient to approve automatic attendance.

2.8.5 Backend Server

The backend will manage:

- Authentication and authorization.
- Staff accounts.
- Institutional configuration.
- Attendance-request validation.
- Attendance-state transitions.
- Duplicate prevention.
- Attendance history.
- Correction requests.
- Reporting.
- Audit records.

The backend will receive location readings from the mobile application. It will not assume that it can directly obtain a phone's current position without communication from the device.

2.8.6 Centralized Database

The database will store accounts, configuration, attendance sessions, attendance requests, supporting event evidence, and administrative changes.

2.8.7 Administrator Web Interface

The web interface will provide authorized access to account management, attendance review, configuration, correction approval, and reporting.

2.9 Backend Database Management

2.9.1 Proposed Database Structure

The prototype will use a relational database with the following main tables.

Table	Main fields and purpose
Users	User ID, staff ID, name, email, department, password hash, role, account status, timestamps
GeofenceConfigurations	Configuration ID, centre latitude, centre longitude, radius, grace period, location-quality thresholds, version, effective time
AttendanceSessions	Session ID, user ID, check-in time, check-out time, method for each action, session status
AttendanceRequests	Request ID, user ID, session ID where applicable, request type, notification time, deadline, status, configuration version
LocationEvidence	Evidence ID, request ID, coordinates, measurement time, reported uncertainty, calculated distance, classification
CorrectionRequests	Correction ID, user ID, related session where available, requested change, reason, review status, reviewer, review time

AuditLogs	Log ID, acting user or system process, action, affected record, previous values, new values, reason, timestamp

2.9.2 Relationships

One user may have multiple attendance sessions, attendance requests, and correction requests.

Each attendance request will reference the configuration version used for its decision. Its supporting location evidence will be linked through the request identifier.

A completed attendance session will retain separate methods for check-in and check-out. For example, a session may contain a manual check-in and an automatic check-out.

2.9.3 Data Integrity

The backend and database will enforce the following rules:

1. Staff identifiers and institutional email addresses must be unique.
2. Each user may have no more than one open attendance session.
3. Check-out cannot occur without an open session.
4. Check-out time cannot precede check-in time.
5. Each request identifier may produce no more than one successful attendance action.
6. Related attendance changes must be committed in a single transaction.
7. Administrative corrections must retain an audit trail.

2.9.4 Corrections and Audit History

A correction will not silently replace the original attendance information.

The system will retain the original values, approved replacement values, correction reason, responsible administrator, and correction timestamp.

Staff members will be able to see whether a record has been corrected.

2.9.5 Configuration Changes

Changes to the institutional centre, radius, grace period, or location-quality thresholds will create a new configuration version.

An already pending attendance request will retain its original configuration version until completion or cancellation. This avoids changing its decision rules partway through the grace period.

2.9.6 Storage and Access Management

The database will be accessed through the backend rather than directly by the mobile application.

The proposed system will:

- Restrict database access to authorized services and administrators.
- Protect communication with encryption.
- Store only location evidence needed for attendance and evaluation.
- Avoid collecting continuous travel history outside the attendance workflow.
- Support backups and restoration testing.
- Apply an institution-approved retention period before deployment.

Timestamps will be stored consistently and displayed in the Asia/Bangkok time zone.

2.10 Failure Handling and Operational Requirements

2.10.1 Network Unavailability

When the network is unavailable, the application may retain event information locally for later synchronization. It will clearly label unsynchronized actions.

An offline action will not be displayed as officially recorded before backend approval. Evidence that no longer satisfies the freshness requirement will be retained for review rather than used to approve delayed automatic attendance.

2.10.2 Location or Notification Unavailability

If the application cannot obtain a valid location reading, it will not automatically record attendance.

If a notification cannot be posted, the system will not treat the user as having ignored a successfully delivered attendance request. It will record the problem and make the unresolved action visible in the application.

2.10.3 Background Execution

The intended workflow requires attendance monitoring while the application is not open on the screen. Background behaviour and timing must be verified on the selected devices during implementation.

The design will not assume that a grace-period timer guarantees execution at an exact moment. Whenever an automatic attempt actually runs, it must obtain valid current evidence before approval.

2.10.4 Usability and Reliability

The interface must clearly distinguish:

- Recorded attendance.
- Pending attendance.
- Uncertain location.
- Failed actions.
- Unsynchronized information.

The system must avoid reporting success when a request has not been accepted by the backend.

Chapter 3: Evaluation Methodology

3.1 Evaluation Objectives

The evaluation will determine whether the prototype can:

1. Classify device location relative to the institutional region.
2. Generate appropriate attendance notifications.
3. Record manual attendance correctly.
4. Record automatic attendance only when the defined conditions are satisfied.
5. Prevent duplicate and contradictory records.
6. Handle unavailable location, notification, and network services.
7. Enforce user permissions.
8. Provide a usable staff and administrator interface.

3.2 Experimental Ground Truth

Testers will follow predefined entry, exit, and boundary scenarios. An observer will record the actual test actions and crossing times.

Ground truth will be established independently of the application's own location classification. The application's GPS output will not also be used as the sole reference for judging whether that output is correct.

Any uncertainty in the reference position or crossing time will be documented. Tests too close to the boundary to establish reliable ground truth will be identified separately.

3.3 Location-Classification Metrics

3.3.1 Confusion Matrix

For readings with a definite Inside or Outside result:

Result	Definition
True Positive (TP)	Actually inside and classified as inside
True Negative (TN)	Actually outside and classified as outside

False Positive (FP)	Actually outside but classified as inside
False Negative (FN)	Actually inside but classified as outside

3.3.2 Accuracy

Accuracy measures the proportion of definite classifications that are correct.

Accuracy (%) = [(TP + TN) / (TP + TN + FP + FN)] × 100

3.3.3 Precision

Precision measures how often an Inside classification is correct.

Precision = TP / (TP + FP)

3.3.4 Recall

Recall measures how many actual Inside cases are correctly classified as Inside.

Recall = TP / (TP + FN)

3.3.5 F1-Score

F1-Score summarizes the balance between precision and recall.

F1-Score = 2 × Precision × Recall / (Precision + Recall)

A metric with a zero denominator will be reported as not applicable.

3.3.6 Uncertain Classification Rate

Uncertain readings will be reported separately rather than silently removed from the evaluation.

Uncertain Rate (%) = Uncertain Readings / Total Evaluated Readings × 100

Classification Coverage (%) = Definite Classifications / Total Evaluated Readings × 100

Coverage will be reported alongside accuracy so that a system producing many Uncertain results is not presented as highly reliable based only on a small number of definite classifications.

3.4 Timing Metrics

3.4.1 Geofence Detection Delay

Detection Delay = Geofence Detection Time − Observed Boundary Crossing Time

Entry and exit delays will be reported separately.

3.4.2 Notification Delay

Notification Delay = Notification Posting Time − Geofence Detection Time

Where notification appearance can be observed during testing, the observed appearance time will also be recorded.

3.4.3 Attendance Recording Delay

Attendance Recording Delay = Backend Approval Time − Observed Boundary Crossing Time

This measurement includes the effect of the grace period for automatic attendance.

Timing results will include the median, maximum, and number of missed events. Missing detections will not be assigned an artificial delay of zero.

3.5 Attendance Metrics

3.5.1 Expected-Action Success Rate

Expected-Action Success Rate (%) = Correctly Recorded Expected Actions / Total Tests Requiring an Attendance Action × 100

A successful action must have the correct user, action type, recording method, and session state, without duplicates.

3.5.2 Automatic Check-In Success Rate

Automatic Check-In Success Rate (%) = Correct Automatic Check-Ins / Total Tests Requiring Automatic Check-In × 100

3.5.3 Automatic Check-Out Success Rate

Automatic Check-Out Success Rate (%) = Correct Automatic Check-Outs / Total Tests

Requiring Automatic Check-Out × 100

3.5.4 False Automatic Attendance Rate

False Automatic Attendance Rate (%) = Incorrect Automatic Actions / Total Recorded Automatic Actions × 100

Incorrect automatic actions include check-in while outside, check-out after returning inside, and actions approved without valid location evidence.

3.5.5 Correct Non-Action Rate

Some tests require the system to avoid recording attendance.

Correct Non-Action Rate (%) = Tests Correctly Producing No Attendance Action / Total Tests Requiring No Attendance Action × 100

This metric will cover temporary entry, leave-and-return behaviour, invalid readings, and duplicate submissions.

3.6 Functional Testing Scenarios

Test	Scenario	Expected result
1	Entry followed by manual confirmation	One manual check-in is recorded
2	Entry with no response; valid Inside reading after timeout	One automatic check-in is recorded
3	Entry followed by confirmed exit before timeout	Pending check-in is cancelled
4	Exit followed by manual confirmation	The open session receives one manual check-out
5	Exit with no response; valid Outside reading after timeout	One automatic check-out is recorded
6	Exit followed by confirmed return before timeout	Pending check-out is cancelled
7	Uncertain location near the boundary	No attendance approval until location is valid
8	Old or low-quality location reading	The action remains unresolved
9	Repeated submission of the same request	No duplicate attendance action

10	Manual and automatic requests arrive together	Only one action succeeds
11	Check-out without an open session	Request is rejected
12	Location permission is unavailable	Monitoring limitation is shown; no automatic action
13	Notification cannot be posted	No ignored-notification automatic action
14	Network connection is lost	Pending synchronization is shown; no false success
15	Application is in the background or phone is locked	Behaviour and delays are measured
16	Staff requests another user's records	Access is rejected
17	Account is deactivated	New attendance submissions are rejected
18	Administrator corrects attendance	Original and corrected values are auditable
19	User logs out with an open session	Session remains open; monitoring-stop warning is shown
20	Configuration changes during a pending request	The request retains its original configuration version

Indoor locations and different available smartphone models will also be tested.

3.7 Test Data Collection

Each test will record:

- Test identifier and scenario.
- Device model and operating-system version.
- Configuration version.
- Institutional radius and grace period.
- Ground-truth location and observed crossing time.
- Geofence detection time.
- Notification posting time.
- Location measurement time and uncertainty.
- Calculated distance and classification.

- User response.
- Backend approval time, if any.
- Attendance method and resulting state.
- Expected and actual outcomes.
- Error category and relevant logs.

3.8 Parameter Comparison

The initial configuration will use a 200-metre radius and a five-minute grace period.

Alternative radii may be tested after inspecting the faculty layout, with all tested radii remaining within the specified 1,000-metre maximum. Alternative grace periods may also be compared.

Where practical, one parameter will be changed at a time. This will help distinguish the effect of radius changes from the effect of grace-period changes.

The selected configuration will consider false attendance, missed attendance, classification coverage, and recording delay. Selection will not rely only on the highest location-classification accuracy.

3.9 Usability Evaluation

Participants will be asked to complete tasks such as logging in, understanding their attendance state, responding to a notification, finding attendance history, and submitting a correction request.

The evaluation will record task completion, misunderstandings, and participant feedback.

Particular attention will be given to whether users understand:

1. The difference between location status and attendance status.
2. When automatic attendance may occur.
3. What dismissing a notification means.
4. Whether an action has been recorded or is still pending.
5. How to report an incorrect record.

Chapter 4: Preliminary Design Status and Technical Challenges

4.1 Current Design Status

The project report defines a proposed attendance workflow and evaluation approach. This revised specification further defines user roles, authentication, interface behaviour, location validation, backend responsibilities, database relationships, and failure handling.

These are design outputs. They do not establish that the system has already been implemented or that its functions have passed testing.

No quantitative experimental results are available in the current report. Accuracy, delay, and attendance-success values will therefore be reported only after testing has been completed.

4.2 Location Uncertainty Near the Boundary

A reported device position near the institutional boundary may not provide enough confidence to classify the device as inside or outside.

The proposed response is to introduce an Uncertain classification and request a fresh reading rather than force an attendance decision.

Testing will determine how often uncertainty occurs and whether the selected radius is suitable for the faculty environment.

4.3 Delayed or Unavailable Background Processing

The application may not complete a location check at the intended timeout.

The system will record actual processing times and validate fresh location evidence whenever the attempt runs. If valid evidence cannot be obtained, the attendance action will remain unresolved.

4.4 Temporary Entry and Return Visits

A user may enter the region briefly or leave and return before the grace period expires.

Pending requests will therefore be cancelled when a confirmed opposite transition occurs.

However, remaining inside a geographic region does not prove an intention to work. A passer-by who remains inside long enough could still satisfy a location-only rule. This limitation must be considered when selecting the institutional boundary and reviewing automatic attendance.

4.5 Duplicate and Conflicting Events

Repeated location events, network retries, and simultaneous manual and automatic actions may produce duplicate requests.

Unique request identifiers, database constraints, and transactional processing will be used to ensure that one attendance request produces no more than one valid action.

4.6 Network and Synchronization Problems

Connectivity failures may delay communication with the backend.

The interface will show the difference between a locally pending event and an approved attendance record. Old location evidence will not be treated as a current position solely because it has just reached the server.

4.7 Privacy and Administrative Accountability

Location information will be limited to what is needed for attendance decisions and evaluation.

Access will depend on user roles. Administrative changes will be recorded with reasons and an audit history. Data-retention rules must be agreed with the institution before operational deployment.

4.8 Development Priorities

The proposed implementation order is:

1. Account management, login, and role enforcement.
2. Database structure and attendance-state validation.
3. Staff and administrator interfaces.
4. Location acquisition and local distance calculation.

5. Manual attendance processing.
6. Notifications and pending-request management.
7. Automatic attendance with fresh location validation.
8. Duplicate prevention and failure handling.
9. Correction requests and audit logs.
10. Field testing and parameter adjustment.

Chapter 5: Expected Outcomes and Project Limitations

5.1 Expected System Deliverables

The project is expected to deliver:

1. A mobile application for authorized staff.
2. A configured institutional attendance region.
3. Check-in and check-out notifications.
4. Manual attendance confirmation.
5. Automatic attendance subject to timeout and location-validation rules.
6. A backend service and centralized attendance database.
7. An administrator interface.
8. Attendance correction and audit functions.
9. Experimental results evaluating functionality, reliability, and usability.

5.2 Expected Staff Workflow

A staff member will log in and complete the required permission setup.

When an eligible entry is detected, the application will request manual check-in. If the user does not respond within the grace period, the system will attempt automatic check-in using a fresh location reading.

When a checked-in user leaves the region, the application will request manual check-out. If the user does not respond, the system will attempt automatic check-out after revalidating that the device is outside.

If the location is uncertain or another required condition is unavailable, the system will leave the action unresolved and display the issue for review.

5.3 Expected Administrative Benefits

The system is intended to reduce forgotten attendance actions, provide centralized attendance records, and make manual and automatic actions distinguishable.

The administrator interface is expected to support account management, attendance review, correction handling, reporting, and investigation of system events.

These are expected benefits. Their extent must be established through implementation and evaluation.

5.4 Project Limitations

The system will depend on users carrying their registered devices and keeping the required services and permissions available.

The circular institutional region will represent an attendance area rather than an exact building boundary. Location readings may be uncertain, and attendance events may be delayed or unavailable.

Automatic attendance will remain a conditional backup mechanism. It will not guarantee that every physical arrival or departure produces an immediate record.

The system will also be unable to establish physical identity or working activity from device location alone.

5.5 Final Evaluation Objective

The final evaluation will determine whether the proposed combination of manual confirmation, location-based reminders, and conditional automatic attendance provides a practical attendance-assistance system for the selected faculty environment.

The conclusion will be based on measured classification performance, attendance outcomes, delays, failure handling, and user feedback.

References

[1] I. Eweoya, O. J. Adeniyi, A. O. Awoniyi, E. Mgbeahuruike, J. O. Adewuyi, T. O. Adigun, and Y. A. Mensah, "Design and Implementation of a University Attendance Management System Using Geo-Fencing," *Asian Journal of Computer Science and Technology*, vol. 14, no. 1, pp. 28–46, 2025. DOI: 10.70112/ajcst-2025.14.1.4323.

[2] "Attendance Management System Using Geofencing Technology," *2024 First International Conference on Pioneering Developments in Computer Science & Digital Technologies (IC2SDT)*, IEEE, 2024. DOI: 10.1109/IC2SDT62152.2024.10696345.

[3] Android Developers, "Create and Monitor Geofences," Android Location and Sensors Documentation.